

STROMTANGEN FYR, Norway

WMO: 014950

Lat: 59.151N

Lon: 10.829E

Elev: 10

StdP: 101.20

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.6	-9.6	-14.6	1.1	-10.7	-12.6	1.3	-8.6	15.7	4.2	15.0	3.5	4.7	40	0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.0	23.8	17.9	22.2	17.5	20.9	17.0	19.3	22.3	18.4	21.3	17.6	20.2	4.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.0	21.0	17.2	12.3	20.2	16.3	11.6	19.3	55.0	22.4	52.2	21.5	49.5	20.3	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	14.4	13.0	DB	-12.3	26.4	2.9	2.1	-14.4	27.9	-16.1	29.1	-17.8	30.3	-19.9	31.8	(4)
(5)			WB	-13.9	20.2	2.7	1.1	-15.8	21.0	-17.4	21.7	-18.9	22.4	-20.8	23.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-0.2	-0.9	1.9	6.4	11.2	15.1	17.7	17.0	13.8	8.9	4.9	1.4		
	DBStd	7.29	4.26	3.96	3.38	2.76	3.13	2.32	2.23	1.85	2.24	2.93	3.71	4.88		
	HDD10.0	1489	317	304	251	115	21	0	0	0	2	56	155	267		
	HDD18.3	3751	576	538	509	359	222	101	40	49	135	292	404	526		
	CDD10.0	813	0	0	0	6	57	153	238	218	117	22	1	0		
	CDD18.3	34	0	0	0	0	1	4	20	9	0	0	0	0		
	CDH23.3	67	0	0	0	0	4	9	47	6	0	0	0	0		
	CDH26.7	3	0	0	0	0	0	1	2	0	0	0	0	0		
(14) Wind	WSAvg	6.2	6.5	5.9	5.8	5.4	5.6	5.9	5.8	6.1	6.6	6.7	6.9	6.8		
(15) Precipitation	PrecAvg	831	68	51	40	47	58	75	74	91	82	101	87	69		
	PrecMax	1055	143	123	134	86	87	146	129	189	204	232	169	140		
	PrecMin	635	8	16	2	14	7	32	28	28	32	12	23	23		
	PrecStd	120	34	28	28	22	22	30	28	39	44	53	35	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	6.9	11.1	17.2	23.1	24.6	26.7	24.2	20.2	15.6	12.2	9.7		
		MCWB	6.9	4.6	6.7	10.3	15.7	16.9	19.4	18.0	14.6	13.0	10.4	8.6		
	2%	DB	7.1	6.0	8.4	14.6	20.2	22.0	24.8	22.7	18.4	14.5	11.3	8.6		
		MCWB	5.7	4.4	4.7	9.6	14.3	16.2	18.8	17.8	14.6	12.4	9.8	7.3		
	5%	DB	6.2	5.0	6.9	12.3	18.0	20.3	22.8	21.1	17.6	13.8	10.5	7.9		
		MCWB	5.1	3.9	4.1	7.9	13.4	16.0	18.2	17.3	14.6	11.9	9.0	6.6		
	10%	DB	5.4	4.2	5.8	10.4	16.0	18.9	21.4	20.0	17.0	13.2	9.8	7.2		
		MCWB	4.2	3.1	3.9	7.1	12.2	14.7	17.7	16.5	14.4	11.4	8.4	5.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	5.4	7.1	10.8	16.9	19.7	20.6	19.5	16.9	14.1	11.2	9.1		
		MCDB	7.2	6.0	10.5	14.9	21.2	21.8	24.3	22.5	18.1	15.0	11.8	9.2		
	2%	WB	5.7	4.6	5.6	9.6	15.4	17.9	19.7	18.6	16.1	13.1	10.2	7.6		
		MCDB	6.3	5.5	7.4	12.9	19.1	21.0	23.2	21.4	17.4	14.1	11.0	8.3		
	5%	WB	5.1	3.9	4.9	8.6	14.2	16.6	18.9	17.7	15.5	12.3	9.3	6.8		
		MCDB	5.9	5.0	6.1	10.9	17.1	19.5	22.0	20.3	17.0	13.5	10.3	7.6		
	10%	WB	4.3	2.9	4.3	7.5	12.9	15.6	18.1	17.1	14.9	11.6	8.5	6.0		
		MCDB	5.2	3.9	5.5	9.6	15.6	18.0	20.8	19.4	16.6	12.9	9.6	7.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	3.6	4.4	5.1	5.1	4.6	4.0	4.1	4.1	4.1	3.5	3.8		
		MCDBR	3.7	3.4	5.9	8.4	8.7	7.2	6.5	6.0	4.7	3.8	3.6	3.2		
	5% WB	MCWBR	3.5	3.1	3.9	4.6	4.7	4.4	3.7	3.3	3.6	3.5	3.7	3.7		
		MCWBR	3.2	2.8	4.8	6.7	7.7	6.1	5.4	4.6	3.7	3.3	3.4	3.2		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.321	0.340	0.379	0.373	0.361	0.382	0.378	0.348	0.336	0.303	0.262		
		taud	2.237	2.318	2.364	2.313	2.408	2.465	2.426	2.432	2.501	2.427	2.288	2.162		
	Ebn at Noon		522	694	798	822	858	874	843	818	789	673	496	407		
		Edh at Noon	44	68	86	107	105	101	103	95	76	61	42	32		
(44) All-Sky Solar Radiation	RadAvg		0.31	0.89	2.39	3.85	5.14	5.86	5.27	4.14	2.75	1.22	0.42	0.21		
	RadStd		0.06	0.16	0.25	0.52	0.49	0.38	0.61	0.28	0.29	0.21	0.10	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page